Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

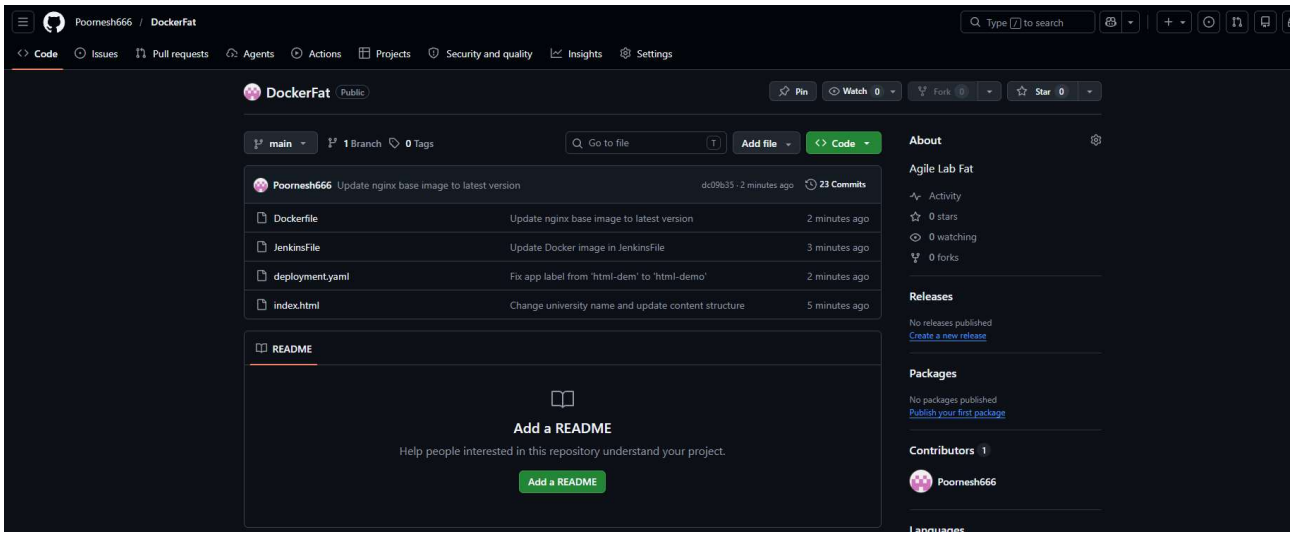
SCHOOL OF COMPUTER SCIENCE ENGINEERING AND INFORMATION SYSTEMS
FINAL ASSESSMENT TEST
WINTER SEMESTER 2025-2026

Programme Name & Branch	**: M.Tech (SE)**
Course Code and Course Name	**: ISWE406P – AGILE DEVELOPMENT PROCESS & DEVOPS LAB**
Faculty Name(s)	**: Prof. Sweta Bhattacharya**
Class Number(s)	**: 2389**
Slot	**: L49+L50**
Date of Examination	**: 16.04.26**

Regd. No: 23MIS0146
Name: Ganji Poornesh Naga Sai Hanuman

Q. Design and develop a University Department Web Page using HTML that provides essential information about a specific department (e.g., Computer Science, Mechanical Engineering). The webpage should include a header section with the university name and its detailed description and administrative heads. The main content should contain an About section describing the department, along with details of courses offered, including course names and brief descriptions. It can also include a Faculty section (optional) listing faculty member with their names, designations, and areas of expertise (basic text is sufficient). Additionally, the page should include a Contact section with details such as address, phone number, and email. A footer section should be added with copyright information. Maintain the source code in a GitHub repository [10].

GitHub repository:



i. Clone the repository into VS Code, make any updates, commit and push the changes to the Git repository [10].

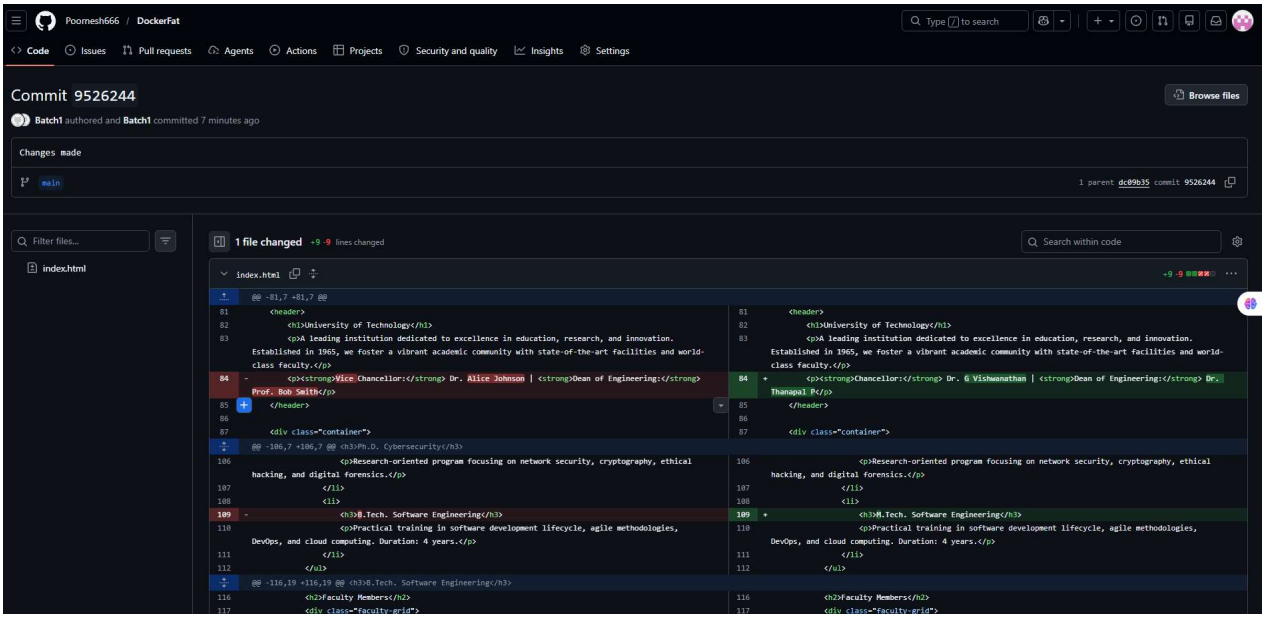
Cloning:

```
PS C:\Users\batch1> git clone https://github.com/Poornesh666/DockerFat.git
Cloning into 'DockerFat'...
remote: Enumerating objects: 58, done.
remote: Counting objects: 100% (58/58), done.
remote: Compressing objects: 100% (54/54), done.
remote: Total 58 (delta 15), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (58/58), 26.88 KiB | 2.69 MiB/s, done.
Resolving deltas: 100% (15/15), done.
PS C:\Users\batch1>
```

Making Changes, Commit & Push:

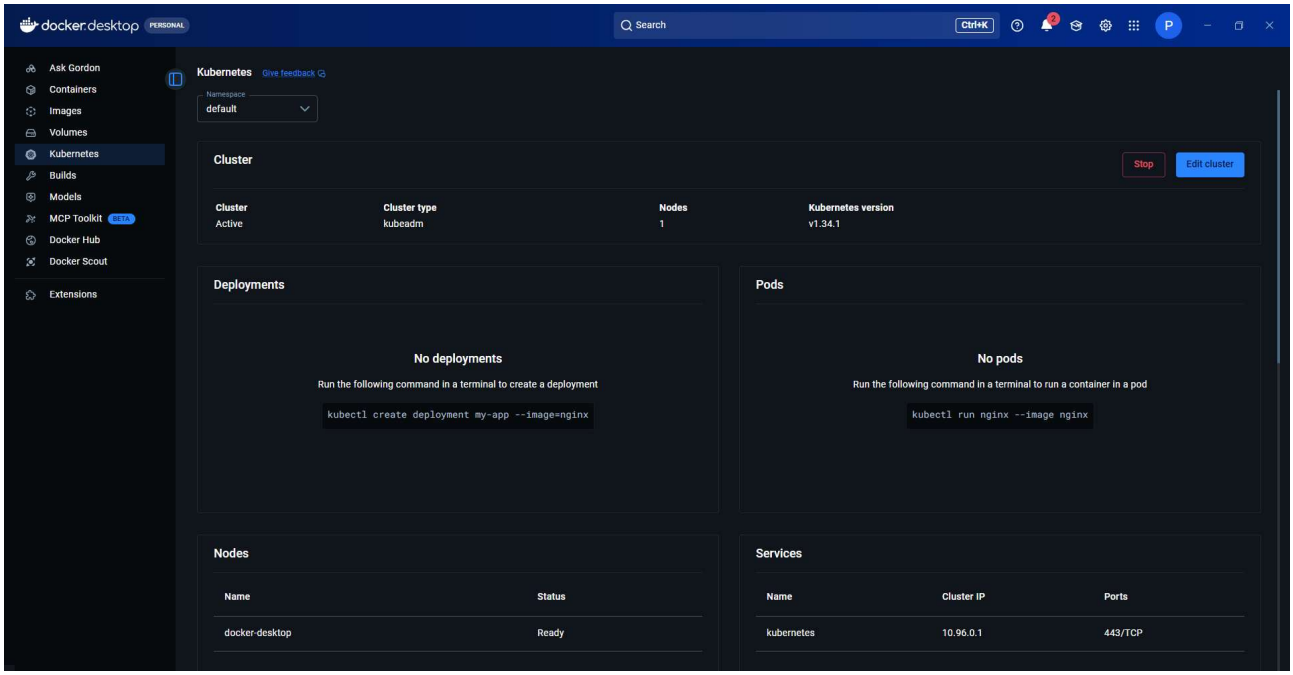
```
PS C:\Users\batch1\Desktop\23MIS0146\DockerFat> git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 24 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 461 bytes | 461.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Poornesh666/DockerFat.git
   dc09b35..9526244  main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\batch1\Desktop\23MIS0146\DockerFat> git add .
PS C:\Users\batch1\Desktop\23MIS0146\DockerFat> git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
PS C:\Users\batch1\Desktop\23MIS0146\DockerFat> git commit -m "Made Changes"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Commit History on GitHub:

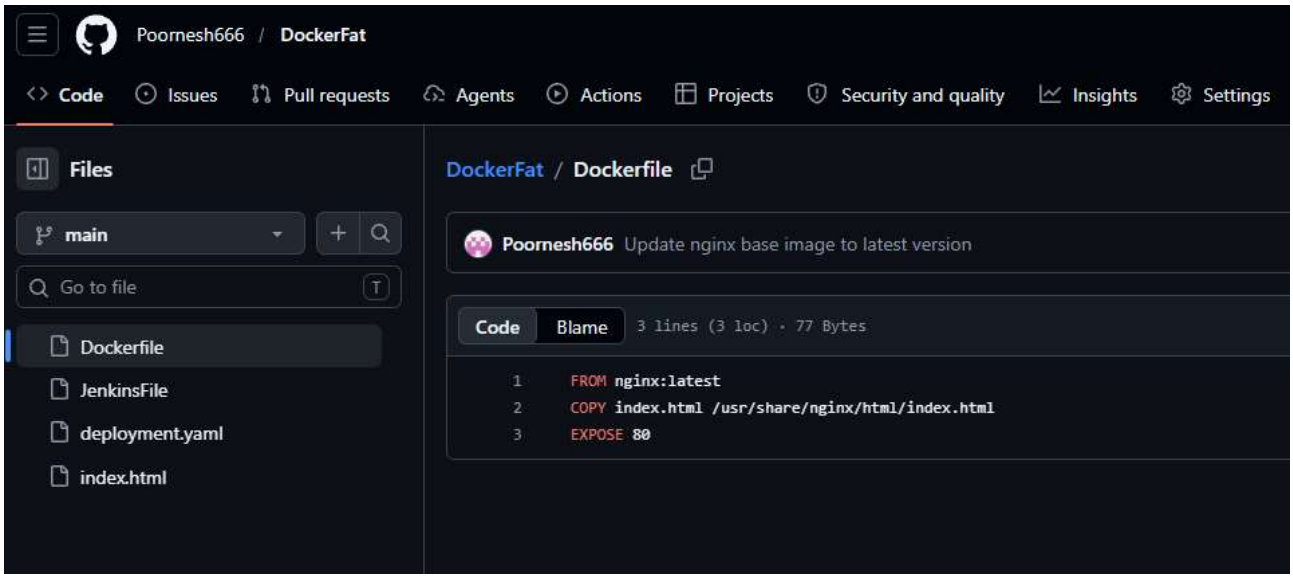


ii. Containerize the application using Docker and prepare it for deployment [10].

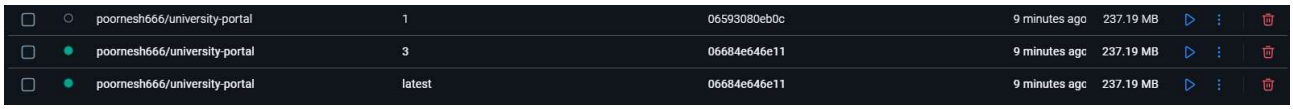
Docker Setup:



DockerFile:



Docker Images:



poornesh666/university-portal	1	06593080eb0c	9 minutes ago	237.19 MB	
poornesh666/university-portal	3	06684e646e11	9 minutes ago	237.19 MB	
poornesh666/university-portal	latest	06684e646e11	9 minutes ago	237.19 MB	

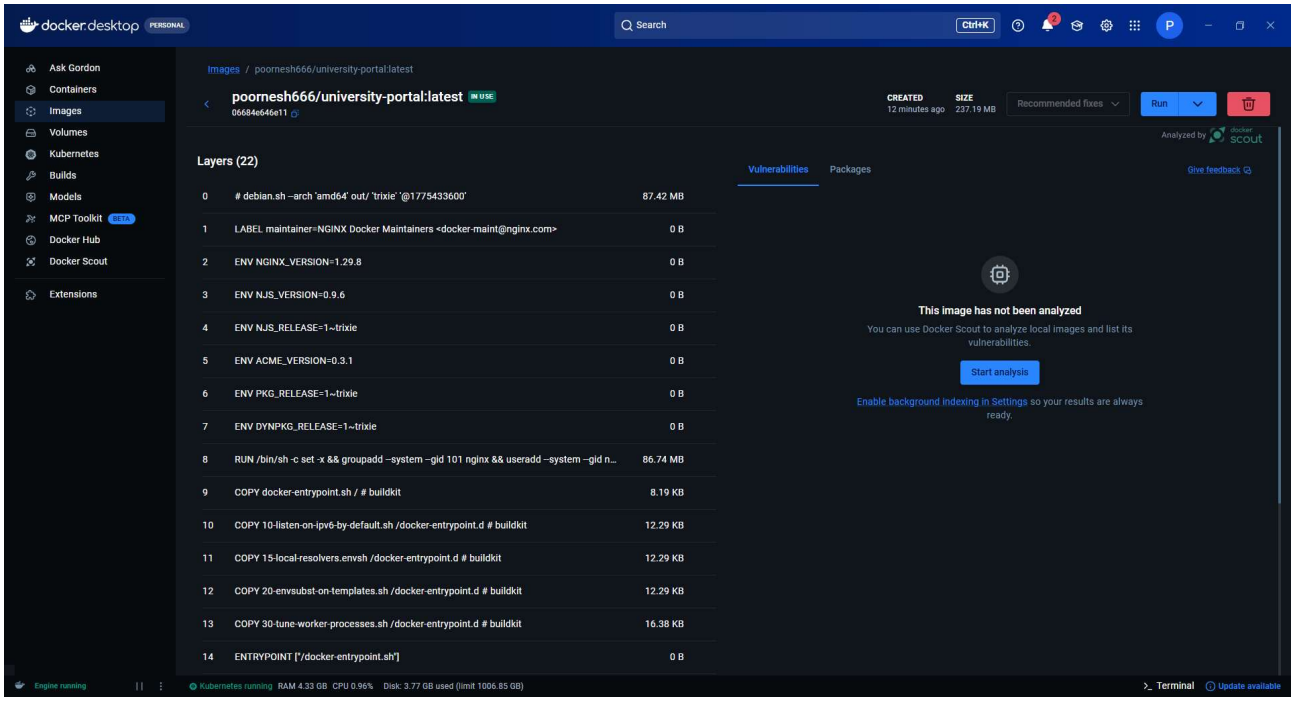
iii. Configure Jenkins to create a pipeline project that builds, tests, and deploys the application using Docker and Kubernetes, and visualize the deployment through Docker Desktop [20].

Jenkins Pipeline:

Console Output:

```
C:\ProgramData\Jenkins\.jenkins\workspace\Lab_Fat>docker tag poornesh666/university-portal:3 poornesh666/university-portal:latest
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\Lab_Fat>docker images --filter=reference="poornesh666/university-portal:*"
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
poornesh666/university-portal:1      06593080eb0c        237MB           63MB
poornesh666/university-portal:3      06684e646e11        237MB           63MB
poornesh666/university-portal:latest 06684e646e11        237MB           63MB
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Push to Docker Hub)
[Pipeline] echo
========== Pushing image to Docker Hub ==========
[Pipeline] script
[Pipeline] {
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\Lab_Fat>echo Logging into Docker Hub...
Logging into Docker Hub...

C:\ProgramData\Jenkins\.jenkins\workspace\Lab_Fat>docker login -u poornesh666 -p ****
WARNING! Using --password via the CLI is insecure. Use --password-stdin.
Login Succeeded

C:\ProgramData\Jenkins\.jenkins\workspace\Lab_Fat>echo Pushing poornesh666/university-portal:3...
Pushing poornesh666/university-portal:3...

C:\ProgramData\Jenkins\.jenkins\workspace\Lab_Fat>docker push poornesh666/university-portal:3
The push refers to repository [docker.io/poornesh666/university-portal]
4a038fd18db1: Waiting
054715a6bffa: Waiting
5435b2dcdf5c: Waiting
```

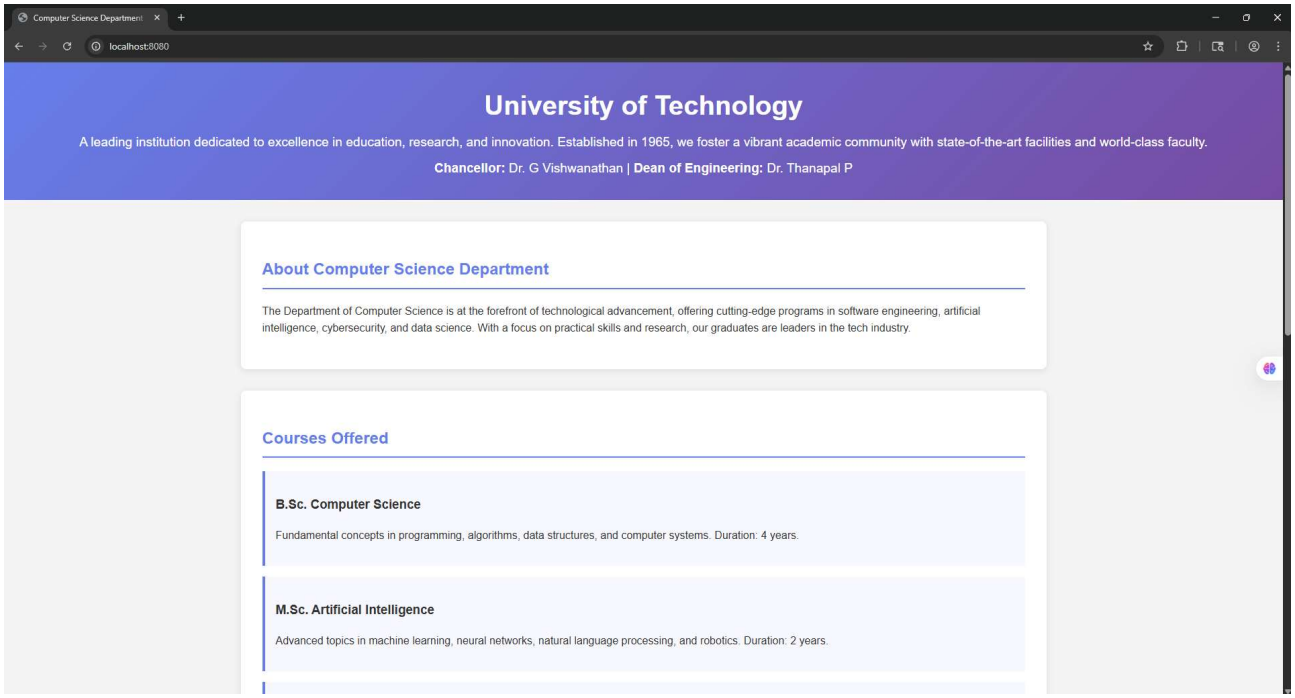
Docker Image Console:



Running Images:

```
C:\Users\batch1>docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED             STATUS                      PORTS     NAMES
6bd08259c32c   poornesh666/university-portal:latest   "/docker-entrypoint.…"   About a minute ago   Up About a minute (healthy)   80/tcp    wizardly_albattani

C:\Users\batch1>docker run -d -p 8080:80 --name web-app poornesh666/university-portal:latest
2d09b94d1367e68988d384450ed247cf0d63e8ca8f2795e3ecf9a0d306d3e7fe
```

Running Website in localhost:8080



Code files:

1. DockerFile:

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

2. JenkinsFile(Pipeline):

```
pipeline {
    agent any
    environment {
        DOCKER_IMAGE = 'poornesh666/university-portal'
        DOCKER_TAG = "${BUILD_NUMBER}"
        DOCKER_CREDENTIALS = credentials('dockerhub')
    }
    stages {
        stage('Checkout') {
            steps {
                echo '========== Checking out code ==========='
                checkout scm
                echo 'Code checked out successfully'
            }
        }
        stage('Build Docker Image') {
            steps {
                echo '========== Building Docker image ==========='
                script {
                    bat 'echo Building image: %DOCKER_IMAGE%:%DOCKER_TAG%'
                    bat 'docker build -t %DOCKER_IMAGE%:%DOCKER_TAG% .'
                    bat 'docker tag %DOCKER_IMAGE%:%DOCKER_TAG% %DOCKER_IMAGE
%:latest'
                    bat 'docker images --filter=reference="%DOCKER_IMAGE%:*"'
                }
            }
        }

        stage('Push to Docker Hub') {
            steps {
                echo '========== Pushing image to Docker Hub ==========='
                script {
                    bat """
                        echo Logging into Docker Hub...
                        docker login -u %DOCKER_CREDENTIALS_USR% -p
%DOCKER_CREDENTIALS_PSW%
                        echo Pushing %DOCKER_IMAGE%:%DOCKER_TAG%...
                        docker push %DOCKER_IMAGE%:%DOCKER_TAG%
                        echo Pushing %DOCKER_IMAGE%:latest...
                        docker push %DOCKER_IMAGE%:latest
                        echo Image pushed successfully
                    """
                }
            }
        }
```

```
        stage('Deploy to Kubernetes') {
            steps {
                echo '========== Deploying to Kubernetes ==========='
                script {
                    // Check if kubectl can connect to cluster
                    def clusterConnected = bat(
                        script: 'kubectl cluster-info',
                        returnStatus: true
                    ) == 0

                    if (clusterConnected) {
                        bat """
                            kubectl set image deployment/html-demo html-demo=%DOCKER_IMAGE
%:latest --record
                            kubectl rollout status deployment/html-demo
                        """
                        echo 'Deployment completed successfully'
                    } else {
                        echo 'WARNING: Kubernetes cluster not connected. Skipping K8s deployment.'
                        echo 'Image is available on Docker Hub: %DOCKER_IMAGE%:latest'
                    }
                }
            }
        }

        stage('Verification') {
            steps {
                echo '========== Verifying deployment ==========='
                script {
                    def clusterConnected = bat(
                        script: 'kubectl cluster-info',
                        returnStatus: true
                    ) == 0

                    if (clusterConnected) {
                        bat """
                            kubectl get deployment html-demo
                            kubectl get pods -l app=html-demo
                        """
                    } else {
                        echo 'Kubernetes not connected. Verifying Docker image instead...'
                        bat 'docker pull %DOCKER_IMAGE%:latest'
                        echo 'Image verified on Docker Hub'
                    }
                }
            }
        }
    }

    post {
        always {
```

```
            echo 'Pipeline execution completed'
        }
        failure {
            echo 'Pipeline failed. Check logs above.'
        }
    }
}
```

3. deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: html-demo
spec:
  replicas: 2
  selector:
    matchLabels:
      app: html-demo
  template:
    metadata:
      labels:
        app: html-demo
    spec:
      containers:
      - name: html-demo
        image: poornesh666/university-portal:latest
        ports:
        - containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
  name: html-demo-service
spec:
  type: NodePort
  selector:
    app: html-demo
  ports:
  - port: 80
    targetPort: 80
    nodePort: 30010
```

WEBSITE LOOKUP:

University of Technology

A leading institution dedicated to excellence in education, research, and innovation. Established in 1965, we foster a vibrant academic community with state-of-the-art facilities and world-class faculty.

Chancellor: Dr. G Vishwanathan | **Dean of Engineering:** Dr. Thanapal P

About Computer Science Department

The Department of Computer Science is at the forefront of technological advancement, offering cutting-edge programs in software engineering, artificial intelligence, cybersecurity, and data science. With a focus on practical skills and research, our graduates are leaders in the tech industry.

Courses Offered

B.Sc. Computer Science

Fundamental concepts in programming, algorithms, data structures, and computer systems. Duration: 4 years.

M.Sc. Artificial Intelligence

Advanced topics in machine learning, neural networks, natural language processing, and robotics. Duration: 2 years.

Ph.D. Cybersecurity

Research-oriented program focusing on network security, cryptography, ethical hacking, and digital forensics.

M.Tech. Software Engineering

Practical training in software development lifecycle, agile methodologies, DevOps, and cloud computing. Duration: 4 years.

Faculty Members

Dr. Swetha B

Professor & Head | Agile & DevOps

Prof. Mangayarakarsi

Associate Professor | Cybersecurity, Networks

Dr. Kritika L B

Professor | Software Engineering, Databases

Prof. Hemalatha

Professor | Algorithms, Data Science

Contact Us

Address: Computer Science Dept., University of Technology, 123 Tech Avenue, Innovation City, 12345

Phone: +91 12345 56798

Email: csdept@ut.edu | **Website:** cs.ut.edu

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4. index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Computer Science Department - University of Technology</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            line-height: 1.6;
            margin: 0;
            padding: 0;
            color: #333;
            background-color: #f4f4f4;
        }
        header {
            background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
            color: white;
            padding: 2rem;
            text-align: center;
        }
        header h1 {
            margin: 0;
            font-size: 2.5rem;
        }
        header p {
            font-size: 1.2rem;
            margin: 0.5rem 0 0;
        }
        .container {
            max-width: 1200px;
            margin: 2rem auto;
            padding: 0 2rem;
        }
        section {
            background: white;
            margin-bottom: 2rem;
            padding: 2rem;
            border-radius: 8px;
            box-shadow: 0 2px 10px rgba(0,0,0,0.1);
        }
        h2 {
            color: #667eea;
            border-bottom: 2px solid #667eea;
            padding-bottom: 0.5rem;
        }
        ul {
            list-style-type: none;
            padding: 0;
        }
        ul li {
            background: #f8f9ff;
            margin: 1rem 0;
            padding: 1rem;
            border-left: 4px solid #667eea;
        }
```

```
        .faculty-grid {
            display: grid;
            grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
            gap: 1rem;
        }
        .faculty-member {
            background: #f8f9ff;
            padding: 1rem;
            border-radius: 5px;
        }
        footer {
            background: #333;
            color: white;
            text-align: center;
            padding: 1rem;
            margin-top: 2rem;
        }
        @media (max-width: 768px) {
            header h1 { font-size: 2rem; }
            .container { padding: 0 1rem; }
        }
    </style>
</head>
<body>
    <header>
        <h1>University of Technology</h1>
        <p>A leading institution dedicated to excellence in education, research, and innovation. Established in 1965, we foster a vibrant academic community with state-of-the-art facilities and world-class faculty.</p>
        <p><strong>Chancellor:</strong> Dr. G Vishwanathan | <strong>Dean of Engineering:</strong> Dr. Thanapal P</p>
    </header>

    <div class="container">
        <section id="about">
            <h2>About Computer Science Department</h2>
            <p>The Department of Computer Science is at the forefront of technological advancement, offering cutting-edge programs in software engineering, artificial intelligence, cybersecurity, and data science. With a focus on practical skills and research, our graduates are leaders in the tech industry.</p>
        </section>

        <section id="courses">
            <h2>Courses Offered</h2>
            <ul>
                <li>
                    <h3>B.Sc. Computer Science</h3>
                    <p>Fundamental concepts in programming, algorithms, data structures, and computer systems. Duration: 4 years.</p>
                </li>
                <li>
                    <h3>M.Sc. Artificial Intelligence</h3>
                    <p>Advanced topics in machine learning, neural networks, natural language processing, and robotics. Duration: 2 years.</p>
                </li>
                <li>
                    <h3>Ph.D. Cybersecurity</h3>
                    <p>Research-oriented program focusing on network security, cryptography, ethical hacking, and digital forensics.</p>
```

```
            </li>
            <li>
                <h3>M.Tech. Software Engineering</h3>
                <p>Practical training in software development lifecycle, agile methodologies, DevOps, and
cloud computing. Duration: 4 years.</p>
            </li>
        </ul>
    </section>

    <section id="faculty">
        <h2>Faculty Members</h2>
        <div class="faculty-grid">
            <div class="faculty-member">
                <h3>Dr. Swetha B</h3>
                <p><strong>Professor &amp; Head</strong> | Agile & DevOps</p>
            </div>
            <div class="faculty-member">
                <h3>Prof. Mangayarakarsi</h3>
                <p><strong>Associate Professor</strong> | Cybersecurity, Networks</p>
            </div>
            <div class="faculty-member">
                <h3>Dr. Kritika L B</h3>
                <p><strong>Professor</strong> | Software Engineering, Databases</p>
            </div>
            <div class="faculty-member">
                <h3>Prof. Hemalatha</h3>
                <p><strong>Professor</strong> | Algorithms, Data Science</p>
            </div>
        </div>
    </section>

    <section id="contact">
        <h2>Contact Us</h2>
        <p><strong>Address:</strong> Computer Science Dept., University of Technology, 123 Tech
Avenue, Innovation City, 12345</p>
        <p><strong>Phone:</strong> +91 12345 56798</p>
        <p><strong>Email:</strong> csdept@ut.edu | <strong>Website:</strong> cs.ut.edu</p>
    </section>
  </div>

  <footer>
    <p>&amp;copy; 2024 University of Technology. All rights reserved. | Designed for Computer Science
Department.</p>
  </footer>
</body>
</html>
```